EV Charger Simulator

Test your EV charging network at any scale — before it goes live. | Zero Hardware. Zero Cost. 5-Minute Setup.

OCPP 1.6J COMPLIANT

THE PROBLEM

- A billing defect across 100 stations generates ~500 incorrect invoices per day
- Hardware test lab: €50,000 – €200,000 (€500 – €5,000 per charger)
- Hardware-based integration tests: 6–12 weeks per release cycle
- 23% of new EV deployments report billing or connectivity defects within 30 days of go-live
- Edge cases (concurrent sessions, resets, network drops) are nearly impossible to reproduce with hardware

THE SOLUTION

- Full OCPP 1.6J compliance — the standard governing 85%+ of the world's 4 million+ public charging points
- All 19 OCPP message types: BootNotification → StopTransaction + every server command
- Real energy metering output: energy (Wh), power (W), voltage (V), current (A)
- Fleet mode: 500 independent virtual chargers on a single laptop in under 10 seconds
- Accelerated time: 60-minute charging session completed in 60 real seconds

FOUR MODES — EVERY TEST SCENARIO COVERED

Mode	What It Does	Best For
boot	Connects, boots, stays live — responds to every dashboard command	Manual QA, remote command testing
session	Full Authorize → Start → MeterValues → Stop in seconds	End-to-end billing validation
multi	All connectors charging simultaneously	Smart charging, load balancing
fleet	10, 50, 100+ independent chargers in parallel	Load testing, CI/CD regression suites

REAL NUMBERS

	Physical Hardware	EV Charger Simulator
Unit cost	€500 – €5,000	**€0**
Test lab setup	Days to weeks	**Under 5 minutes**
Concurrent chargers	Limited by budget	**500+ on a single laptop**
Session duration	Real time (60 min = 60 min)	**60 min = 60 seconds**
Edge case scripting	Manual, unreliable	**100% repeatable**
CI/CD integration	Not possible	**Native — GH Actions, GitLab, Jenkins**

GET RUNNING IN UNDER 5 MINUTES

```
npm install  &&  cp config.example.json config.json  &&  npm run dev boot
# Stress-test: EV_SIM_FLEET_COUNT=50 EV_SIM_ACCELERATED=true npm run dev fleet
```